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COMPARISON OF ESTIMATION METHODS FOR GENERALIZED EXTREME VALUE DISTRIBUTION.

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Declaration

We hereby declare that this project entitled, "COMPARISON OF ESTIMATION METHODS FOR GENERALIZED EXTREME VALUE DISTRIBUTION" is our own original work towards the award of the BSc. Statistics degree and that, to the best of my knowledge, it contains no material previously published by another person nor material which had been accepted for the award of any other degree of the university, except where due acknowledgment had been made in the text.

Dedication

We dedicate this paper to God Almighty and to the pillars of support and inspiration. This thesis stands as a testament to the collective endeavor and steadfast encouragement provided by those who have guided and uplifted us throughout this journey.

We dedicate this study to our families who stayed with us all the time with their unconditional love and support. Their strong belief in our opportunity to succeed and willingness to sacrifice many things for us laid the foundation of this accomplishment. Their understanding and readiness to help us in quiet times even during our hardest moments allowed us to continue working on this study.

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Abstract

The Generalized Extreme Value (GEV) distribution is a key element in statistical extreme value theory that encapsulates all the Gumbel, Fréchet, and Weibull distributions in a single parametric model. The value of the shape parameter ξ has a significant impact on the tail behaviour of the distribution as well as on the existence of moments. This paper presents a new Monte Carlo simulation strategy that compares seven estimation techniques, which include Maximum Likelihood Estimates (MLE), ProbabilityWeighted Moments (PWM), LMoments (LMOM), LH-Moments (LHMOM), Method of Moments (MOM), Least Squares (LS) and Weighted Least Squares(WLS) in three different GEV classes (Fréchet with $\xi = 0.15$, Gumbel with $\xi = 0$, and Weibull with $\xi = -0.15$) and seven sample sizes ($n \in \{10, 20, 50, 100, 300, 500, 1000\}$). Performance is assessed using Bias and Root Mean Squared Error (RMSE) over $R = 1000$ Monte Carlo replications. Results show that MLE achieves the lowest RMSE for moderate-to-large samples across all sub-families, while MOM and moment-based methods exhibit systematic bias that does not diminish with increasing sample size in the heavy- and light-tailed cases.

Keywords: Generalized Extreme Value distribution; Methods of Moments; Least Squares; Weighted Least Squares; Linear Hybrid Moments; Maximum Likelihood; Linear-Moments; Probability-Weighted Moments; Monte Carlo simulation; Shape parameter estimation.

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Abbreviations

GEV —- Generalized Extreme Value

μ — Location(Population mean)

σ — Population Standard deviation (scale)

σ^2 — Population Variance

ξ — Shape Parameter

MOM — Method of Moments

MLE — Maximum Likelihood Estimation

PWM — Probability Weighted Moments

LMOM — Linear Moments

LHMOM — Linear Hybrid Moments

LS — Least Squares

WLS — Weighted Least Squares

RMSE — Root Mean Square Error

R — Number of Replications

n —- sample size

$F(x)$— Cumulative density function

$G(x)$ — Cumulative density Function of the GEV distribution

$f(x)$ — Probability Density Function

$\bar{x}$ — sample Mean

S^2 — Sample Variance

C_s — Skewness Coefficient

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Chapter 1

Introduction

1.1 Background of the Study

Extreme events refer to phenomena that are substantially more severe than typical observations, and they pose significant risks across a wide range of domains, including hydrology, climatology, environmental science, finance, and engineering. Events such as floods, intense heat waves, strong wind storms, and heavy rainfall may result in catastrophic loss of life and substantial property damage. A central step toward understanding these events and improving preparedness is the use of appropriate statistical models to characterize their magnitude and frequency.

The Generalized Extreme Value (GEV) distribution offers the conceptual basis for modeling block maxima as the largest values over some time period. The GEV distribution is based on extreme value theory unifying three classical families of extreme values under a single parametric model: Gumbel (light-tailed, $\xi = 0$), Fréchet (heavy-tailed, $\xi > 0$) and Weibull (bounded-tail, $\xi < 0$) distributions (Jenkinson, 1955; Coles, 2001). A random variable X has an extreme value distribution if its GEV representation is given by $X \sim \text{GEV}(\mu, \sigma, \xi)$, where μ, σ and ξ are location, scale and shape parameters respectively.

Of special importance among the three GEV parameters is the shape parameter ξ which determines tail behavior for purposes of extrapolation to extreme quantiles popularly referred to

as return levels. The challenge in obtaining reasonably accurate estimates of ξ has motivated studies comparing relative efficiencies of different parameter estimation procedures especially under conditions of limited data availability.

Statistical literature describes several estimation approaches for fitting GEV distribution that possess different theoretical properties and practical benefits. These include Method of Moments (MOM), Maximum Likelihood Estimation (MLE), Probability Weighted Moments (PWM), Linear Moments (LMOM), Linear Hybrid Moments (LHMOM), Least Squares (LS) and Weighted Least Squares (WLS). However, relative performance among these methods is not always identical; their efficiency highly depends on sample size at hand as well as tail behavior characteristics of underlying distribution. Thus, this study was conducted at Kwame Nkrumah University of Science and Technology, Department of Statistics and Actuarial Science, to compare all seven estimation methods over a wide range of sample sizes and GEV tail conditions via simulation study.

1.2 Problem Statement

The GEV distribution has been widely used to model extreme events in various fields such as hydrology, climatology, and environmental science. Accurate estimation of its parameters is crucial for effective statistical inference, especially when it comes to estimating extreme quantiles like return levels that are often used in policy decisions and risk management strategies. However, the estimation of GEV parameters particularly the shape parameter ξ poses significant statistical challenges when sample sizes are small, which is frequently the case in practice.

Although several estimation methods can be applied and have been individually examined within the literature, an empirical comparison of seven different methods has not been conducted across a wide variation in sample sizes and distributional tail behavior. Often practitioners tend to use Maximum Likelihood Estimation (MLE) and L-Moments (LMOM) without any evidence-based understanding of when to use other methods. The differences among the methods of Probability Weighted Moments (PWM), Linear Hybrid Moments (LHMOM), Least Squares (LS), and Weighted Least Squares (WLS) as compared with these common methods have not been studied using a controlled Monte Carlo simulation for all types of GEV tail behaviors (Fréchet, Gumbel, Weibull). Not knowing this may lead to the use of inefficient estimators in practice resulting in poor estimates of return levels and risk assessments.

This paper aims to fill this gap in the literature by comparing the number of techniques for estimating GEV parameters using a Monte Carlo simulation with different sample sizes $n \in \{10, 20, 50, 100, 300, 500, 1000\}$ for three tail conditions and applying the idea to real data (Carcassonne heat wave).

1.3 Objectives of the Study

1.3.1 Main Objective

The main objective of this study is to compare the performances of seven parameter estimation techniques for the GEV distribution through an objective and systematic assessment over a wide range of sample sizes and tail behaviors and further illustrate the application of the best technique on actual climatic data.

1.3.2 Specific Objectives

The specific objectives of this research are:

1. Comparing the performance of the Seven estimation methods using their Root Mean Square Error.

2. To assess the consistency of estimation methods across sample sizes under each tail condition.

3. To use the recommended method on real climate data (Carcassonne heat wave data) in order to compute GEV return levels with their corresponding 95% confidence intervals.

1.4 Research Questions

The study intends to answer the research questions mentioned below, based on the objectives stated above.

Main Research Question

What is the best and most accurate parameter estimation procedure for estimating the GEV shape parameter over various sample sizes and tail behaviors?

Specific Research Questions

1. What is the RMSE of the shape parameter ξ under all seven estimation methods.(Roslan et al., 2020)?

2. How does each estimation method perform with increasing sample size from $n = 10$ to $n = 1000$?

3. Which estimation technique gives consistently least RMSE and close-to-zero bias for each GEV tail type?

4. What is the GEV-based return levels of Carcassonne heat wave data, and how do 95% confidence intervals change when extended to a return period of 2, 10, 20, and 100 years?

1.5 Significance of the Study

The current research results have very broad theoretical and practical implications. From a theoretical viewpoint, it adds to debates in literature about finite sample performance of GEV estimators by extending prior Monte Carlo studies' scope to seven simultaneous estimation methods with larger sample sizes than those considered by Roslan et al. (2020). Results from this simulation provide an empirical basis that is scientifically objective for comparing these methods under identical conditions so as to update current recommendations in extreme value literature.

In concrete terms, evidence-based recommendations on which estimator to choose may be directly beneficial for scientists and professionals working in hydrology, climatology, actuarial science as well as risk management. Practically speaking, since the number of extreme observations is usually small, choice of estimation method can significantly affect reliability of return level estimates. This study has provided clear data-driven answers: MOM is best for very small samples; MLE better when large samples are available; MLE with LS or WLS should be used if moderate sample sizes are given guidance that can immediately apply to extreme value analysis.

The heat wave data from Carcassonne has also shown real-world relevance by applying this methodology as one might measure risk related to temperature extremes through GEV return levels. Due to climate change impacts on the frequency and intensity of extreme weather phenomena occurring today more than ever before, accurate statistical tools for measuring such risks become important. Lastly, this analysis revealed deficiencies in PWM, LMOM, and LHMOM when applied under heavy and bounded tail distributions so that practitioners will not use them indiscriminately.

1.6 Scope and Limitations of the Study

1.6.1 Scope

The present study is confined to seven methodologies for GEV parameter estimation: MOM, MLE, PWM, LMOM, LHMOM, LS, and WLS. The simulation experiment encompasses three canonical GEV scenarios: the Fréchet case (GEV$(0, 1, 0.15)$, heavy tail), the Gumbel case (GEV$(0, 1, 0)$, light tail), and the Weibull case (GEV$(0, 1, -0.15)$, bounded tail). Seven sample sizes $n = 10, 20, 50, 100, 300, 500,$ and $1{,}000$ with $B = 1{,}000$ Monte Carlo replications per scenario are utilized, consistent with the simulation design of Roslan et al. (2020). Performance evaluation is based solely on bias and RMSE of shape parameter ξ. The real data application utilizes Carcassonne Heat data from the `extRemes` package in R, which contains annual heat wave magnitude indices for Carcassonne, France.

1.6.2 Limitations

The limitations of this study are as follows:

1. In the simulation study, only shape parameter ξ is considered; bias and RMSE for location (μ) and scale (σ) parameters fall outside of this work's scope.

2. Only finite GEV cases will be considered here corresponding to values of ξ at -0.15, 0 and 0.15 as per Roslan et al. (2020).

3. The extreme behavior for estimation techniques at very high values of the shape parameter such as greater than 0.5 will not be discussed in this study.

4. This study assumes perfect fitting with GEV distribution whereas practically data can be non-stationary, dependent, or subject to model misspecification; these factors are not included in the simulation framework.

5. Bayesian estimation methods have been excluded from comparison since they have been adopted in extreme value literature for their ability to incorporate prior information.

1.7 Organization of the Study

This paper continues with:

Chapter One which details an introduction to this study comprising background information, problem statement, objectives, research questions, significance, scope and limitations, plus organization of this work.

Chapter Two contains a review of literature regarding GEV distribution concerning its application toward extreme value analysis alongside seven different estimation methods that form part of this study, thus situating current work within available knowledge.

Chapter Three presents the results of the simulation study with descriptive tables of bias and RMSE for all seven methods across all three GEV scenarios and sample sizes considered. It also includes results from the real-data application, including estimated GEV parameters and return levels with confidence intervals.

Chapter Four is the last chapter, which summarizes major findings and conclusions based on both simulation studies and real-data analyses plus practical recommendations for choosing GEV estimation methods. This chapter will also outline potential future research directions.

Chapter 2

Methodology

2.1 Introduction

This chapter presents the methodological framework adopted to compare seven parameter estimation techniques for the Generalized Extreme Value (GEV) distribution, namely the Method of Moments (MOM), Maximum Likelihood Estimation (MLE), Probability Weighted Moments (PWM), L-Moments (LMOM), Linearized High-Order Moments (LHMOM), Least Squares (LS), and Weighted Least Squares (WLS). Each method is described in detail, with its defining equations and derivation steps laid out to facilitate replication and comparison. The evaluation is simulation-based, covering a wide range of sample sizes to capture each method's small and large sample behaviour.

2.2 Research Design

A quantitative, simulation-based comparative design was adopted for this study. Working under controlled conditions made it possible to assess each estimator's bias and root mean square error against known true parameter values, without the confounding effects of real data noise or non-stationary. This design follows the approach of (Roslan, Na and Gabda, 2020), who used Monte Carlo simulation to evaluate MLE, PWM and penalised MLE for small GEV samples, and is consistent with the broader simulation framework described by (Anghel and Ianculescu, 2025)

for seven estimation methods applied to flood frequency data.

2.3 Simulation Design

We made up datasets using the GEV distribution, setting the parameter values ourselves beforehand. Doing this directly from the GEV means we know the exact right answers, which lets us precisely measure how much our estimates are wrong. We chose the numbers for our simulation based on what (Coles, 2001) and (Katz, Parlange and Naveau, 2002) found. Their work showed us the typical ranges for the GEV shape, scale, and location parameters when dealing with extreme weather and water events.

2.3.1 Data Generation and Estimation Procedure

Six sample sizes were considered: $n \in \{10,\ 20,\ 50,\ 100,\ 300,\ 500,\ 1000\}$. Small values ($n = 10, 20, 50$) test each estimator in the regime where bias is known to be most severe, while larger values ($n = 300, 500, 1000$) allow the asymptotic properties of each method to manifest. Each combination of sample size and estimation method was replicated $B = 1{,}000$ times so that the Monte Carlo standard error on the performance statistics is negligible, following the same replication scheme adopted by (Roslan, Na and Gabda, 2020).

All data were generated in RStudio. The `evd` package was used for simulating GEV random variables and for fitting models by MLE, while the `lmom` package provided the L-moment calculations. Methods not directly supported by these packages (LHMOM, LS, WLS) were implemented in custom R scripts following the equations given in (Anghel and Ianculescu, 2025).

2.4 The GEV Distribution

The GEV distribution unifies three classical extreme value families Gumbel, Fréchet and Weibull into a single parametric form, which emerges naturally from the Fisher Tippett Gnedenko theorem on the limiting distribution of normalised block maxima (Coles, 2001; Araveeporn and

Sukpan, 2025). If $X_1, \ldots, X_n$ are independent and identically distributed with common distribution function F, and $M_n = \max\{X_1, \ldots, X_n\}$, then under suitable regularity conditions the normalised maximum converges in distribution:

$$\Pr\left\{ \frac{M_n - b_n}{a_n} \leq x \right\} \longrightarrow G(x), \tag{2.1}$$

where $a_n > 0$ and b_n are norming constants and G belongs to the GEV family (Pinheiro and Grotjahn, 2014). The cumulative distribution function (CDF) of the GEV is

$$F(x;\, \mu, \sigma, \xi) = \begin{cases} \exp\left\{ -\left[1 + \xi \left(\dfrac{x - \mu}{\sigma} \right) \right]^{-1/\xi} \right\}, & \xi \neq 0, \\ \exp\left\{ -\exp\left(-\dfrac{x - \mu}{\sigma} \right) \right\}, & \xi = 0, \end{cases} \tag{2.2}$$

defined on the set $\{x : 1 + \xi(x - \mu)/\sigma > 0\}$, with $\mu \in \mathbb{R}$ (location), $\sigma > 0$ (scale) and $\xi \in \mathbb{R}$ (shape) (Roslan, Na and Gabda, 2020; Anghel and Ianculescu, 2025). When $\xi > 0$ the distribution is of Fréchet type with a heavy upper tail; when $\xi < 0$ it is of Weibull type with a finite upper endpoint; and in the limit $\xi \to 0$ it reduces to the Gumbel (Type I) distribution.

The corresponding probability density function (PDF) for $\xi \neq 0$ is

$$f(x;\, \mu, \sigma, \xi) = \frac{1}{\sigma} \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-(1/\xi)-1} \exp\left\{ -\left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-1/\xi} \right\}, \tag{2.3}$$

and the quantile function is

$$x(p;\, \mu, \sigma, \xi) = \begin{cases} \mu - \dfrac{\sigma}{\xi} \left[(-\ln p)^{-\xi} - 1 \right], & \xi \neq 0, \\ \mu - \sigma \ln(-\ln p), & \xi = 0. \end{cases} \tag{2.4}$$

2.5 Data Analysis

Seven methods were used to estimate the GEV parameters $\boldsymbol{\theta} = (\mu, \sigma, \xi)$. Each is presented below with its theoretical background, key equations and estimation procedure. Performance

is measured by Bias and RMSE as defined in Section 2.6.

2.5.1 Method of Moments (MOM)

The Method of Moments is one of the oldest estimation strategies in statistics, tracing back to Karl Pearson's work at the turn of the twentieth century (Abdulali et al., 2022). It operates by equating the first k theoretical moments of the assumed distribution to the corresponding sample moments and then solving for the k unknown parameters (Abdulali et al., 2022; Anghel and Ianculescu, 2025). For a three-parameter distribution such as the GEV, three equations are required.

Theoretical Moments of the GEV

For $\xi < 1$ (so that the mean exists) and $\xi < 1/2$ (variance exists), the theoretical mean, variance and coefficient of skewness of the $\mathrm{GEV}(\mu, \sigma, \xi)$ distribution are (Anghel and Ianculescu, 2025; Abdulali et al., 2022):

$$\sigma^2 = \frac{\sigma^2}{\xi^2}\left(\Gamma(1-2\xi) - \Gamma(1-\xi)^2\right), \quad \text{valid for } \xi < \tfrac{1}{2} \tag{2.5}$$

$$C_s = \frac{\Gamma(1-3\xi) - 3\Gamma(1-\xi)\Gamma(1-2\xi) + 2\Gamma(1-\xi)^3}{\left(\Gamma(1-2\xi) - \Gamma(1-\xi)^2\right)^{3/2}}, \quad \text{valid for } \xi < \tfrac{1}{3} \tag{2.6}$$

$$\mu_1 = \mu + \gamma_E \sigma, \quad \sigma^2 = \frac{\pi^2}{6}\sigma^2, \quad C_s \approx 1.1396 \tag{2.7}$$

$$\gamma_E \approx 0.5772 \tag{2.8}$$

where $\gamma(\cdot)$ is the Euler Gamma function. For the Gumbel sub-case ($\xi = 0$) these reduce to $\mu_1 = \mu + \gamma_E\sigma$, $\sigma^2 = \pi^2\sigma^2/6$ and $C_s \approx 1.1396$, where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant.

Estimation Procedure

The sample mean $\bar{x}$, standard deviation S and skewness coefficient C_s are computed first:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad Cs = \frac{n}{(n-1)(n-2)S^3}\sum i = 1^n(x_i - \bar{x})^3. \tag{2.9}$$

The shape parameter is then obtained by inverting equation (**??**) with $C_s = C_s$. Because the exact inversion of (**??**) requires iterative numerical solution, (Anghel and Ianculescu, 2025) proposed the following rational approximation, valid for $0.01 < C_s < 6$:

$$\xi = -\frac{0.279530254 - 0.244456673\, C_s}{1 + 0.332978277\, C_s + 0.075850709\, C_s^2 - 0.004873010\, C_s^3}. \tag{2.10}$$

With ξ in hand, the scale and location estimates follow directly from (**??**) and (**??**):

$$\sigma = \frac{|\xi|\, S}{\sqrt{(1-2\xi)^{-2}\,(1-\xi)}}, \tag{2.11}$$

$$\mu = \bar{x} - \frac{\sigma}{\xi}\big[(1-\xi) - 1\big]. \tag{2.12}$$

A notable limitation of MOM is that corrections for higher-order indicators such as skewness and kurtosis can introduce substantial uncertainty when the sample size is small, which partly explains why the method has seen declining use in the regionalisation of extreme events (Anghel and Ianculescu, 2025).

2.5.2 Maximum Likelihood Estimation (MLE)

MLE is the most widely used technique for GEV parameter estimation because it enjoys well-established asymptotic properties consistency, efficiency and asymptotic normality for $\xi > -\frac{1}{2}$ (Roslan, Na and Gabda, 2020; Araveeporn and Sukpan, 2025). Its core idea is to choose

the parameter vector that makes the observed data most probable under the assumed model (Abdulali et al., 2022).

Likelihood Function

For an independent sample $\mathbf{x} = (x_1, \ldots, x_n)$, the likelihood is the product of the GEV densities evaluated at each observation. For $\xi \neq 0$:

$$L(\mu, \sigma, \xi \mid \mathbf{x}) = \frac{1}{\sigma^n} \prod_{i=1}^{n} \left[1 + \xi\left(\frac{x_i - \mu}{\sigma}\right)\right]^{-(1/\xi)-1} \exp\left\{-\sum_{i=1}^{n} \left[1 + \xi\left(\frac{x_i - \mu}{\sigma}\right)\right]^{-1/\xi}\right\}, \tag{2.13}$$

subject to $1 + \xi(x_i - \mu)/\sigma > 0$ for all i (Anghel and Ianculescu, 2025). In practice, the log-likelihood

$$\ell(\mu, \sigma, \xi) = -n \ln \sigma - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{n} \ln\left[1 + \xi\left(\frac{x_i - \mu}{\sigma}\right)\right] - \sum_{i=1}^{n} \left[1 + \xi\left(\frac{x_i - \mu}{\sigma}\right)\right]^{-1/\xi} \tag{2.14}$$

is maximised because the product in (2.13) becomes numerically intractable for large n (Araveeporn and Sukpan, 2025).

Score Equations

Setting the partial derivatives of ℓ to zero produces the three score equations (Araveeporn and Sukpan, 2025; Hossain et al., 2021):

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\xi\sigma} \sum_{i=1}^{n} \left(1 + \xi \frac{x_i - \mu}{\sigma}\right)^{-1} - \frac{1}{\sigma} \sum_{i=1}^{n} \left(1 + \xi \frac{x_i - \mu}{\sigma}\right)^{-(1+1/\xi)} = 0, \tag{2.15}$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma}\left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{n} \frac{(x_i - \mu)/\sigma}{1 + \xi(x_i - \mu)/\sigma} - \frac{1}{\sigma} \sum_{i=1}^{n} \frac{(x_i - \mu)/\sigma}{[1 + \xi(x_i - \mu)/\sigma]^{1+1/\xi}} = 0, \tag{2.16}$$

$$\frac{\partial \ell}{\partial \xi} = \sum_{i=1}^{n} \frac{(x_i - \mu)/\sigma}{1 + \xi(x_i - \mu)/\sigma} \left(\frac{1}{\xi^2} \ln\left[1 + \xi \frac{x_i - \mu}{\sigma}\right] - \frac{1}{\xi}\right) - \sum_{i=1}^{n} \frac{\ln[1 + \xi(x_i - \mu)/\sigma]\,/\xi^2}{[1 + \xi(x_i - \mu)/\sigma]^{1/\xi}} = 0. \tag{2.17}$$

These equations have no closed-form solution because of the nonlinearity introduced by ξ, so estimates are obtained iteratively via the Newton–Raphson method (Araveeporn and Sukpan,

2025):

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \left[\mathcal{H}(\boldsymbol{\theta}^{(t)})\right]^{-1} \nabla \ell\big(\boldsymbol{\theta}^{(t)}\big), \tag{2.18}$$

where $\mathcal{H}$ is the matrix of second-order partial derivatives of ℓ (the Hessian), and iteration continues until $\|\boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^{(t)}\| < 10^{-6}$. As noted by Roslan, Na and Gabda (2020), MLE performs poorly for $n \lesssim 50$ because the shape parameter estimate can have very large variance in small samples, a limitation that motivated the development of PWM and penalised variants.

2.5.3 Probability Weighted Moments (PWM)

Probability weighted moments were introduced by Greenwood et al. (1979) as a generalisation of ordinary moments that incorporates information about the order of the data. Hosking, Wallis and Wood (1985) subsequently derived explicit PWM estimators for the GEV and showed via simulation that they outperform MLE in small to moderate samples a finding corroborated by Roslan, Na and Gabda (2020) and Anghel and Ianculescu (2025).

2.5.4 Definition and Population Values

For a random variable X with CDF F, the PWMs are defined as

$$M_{p,r,s} = [X^p\, F(X)^r\, \{1 - F(X)\}^s] = \int_0^1 \{x(F)\}^p\, F^r\, (1-F)^s\, \mathrm{d}F, \tag{2.19}$$

where $p, r, s \in \mathbb{R}$ (Hosking, Wallis and Wood, 1985). The sub-family $\beta_r = M_{1,r,0} = [X\, F(X)^r]$ is the most useful for the GEV. For the GEV parameterised as $F(x) = \exp\{-[1 - k(x - \xi)/\alpha]^{1/k}\}$ (in the notation of Hosking, Wallis and Wood 1985, where $k = -\xi_{\text{shape}}$, $\alpha = \sigma$, $\xi_{\text{loc}} = \mu$), the population PWMs are

$$\beta_r = (r+1)^{-1}\left\{\xi_{\text{loc}} + \frac{\alpha}{k}\left[1 - (r+1)^{-k}(1+k)\right]\right\}, \quad k > -1. \tag{2.20}$$

In particular, setting $r = 0, 1, 2$ and rearranging:

$$\beta_0 = \xi_{\text{loc}} + \frac{\alpha}{k}\big[1 - (1+k)\big], \tag{2.21}$$

$$2\beta_1 - \beta_0 = \frac{\alpha\,(1+k)\,(1-2^{-k})}{k}, \tag{2.22}$$

$$\frac{3\beta_2 - \beta_0}{2\beta_1 - \beta_0} = \frac{1-3^{-k}}{1-2^{-k}}. \tag{2.23}$$

2.5.5 Sample Estimators and Parameter Estimation

Given the ordered sample $x_{(1)} \le \cdots \le x_{(n)}$, an unbiased estimator of β_r is (Hosking, Wallis and Wood, 1985):

$$b_r = \frac{1}{n}\sum_{j=r+1}^{n} \frac{(j-1)(j-2)\cdots(j-r)}{(n-1)(n-2)\cdots(n-r)}\, x_{(j)}. \tag{2.24}$$

The shape parameter is recovered from (2.23) by replacing β_r with b_r. Because the function $(1-3^{-k})/(1-2^{-k})$ is nearly linear over the range $-\tfrac{1}{2} < k < \tfrac{1}{2}$ typically encountered in practice, Hosking, Wallis and Wood (1985) proposed the low-order polynomial approximation

$$k = 7.8590\,c + 2.9554\,c^2, \qquad c = \frac{2b_1 - b_0}{3b_2 - b_0} - \frac{\ln 2}{\ln 3}, \tag{2.25}$$

with error less than 0.0009 throughout $-\tfrac{1}{2} < k < \tfrac{1}{2}$. Scale and location are then estimated from (2.22) and (2.21):

$$\alpha = \frac{(2b_1 - b_0)\,k}{(1+k)(1-2^{-k})}, \tag{2.26}$$

$$\xi_{\text{loc}} = b_0 + \alpha\big\{(1+k) - 1\big\}/k. \tag{2.27}$$

These estimators satisfy a feasibility condition—$k > -1$ and $\alpha > 0$—almost surely, which guarantees that the estimated distribution is proper (Hosking, Wallis and Wood, 1985).

2.5.6 L-Moments (LMOM)

L-moments, introduced by Hosking (1990), are linear combinations of order statistics that summarise a distribution's location, scale and shape. They share the favourable asymptotic properties of PWMs while being somewhat easier to interpret, and are substantially less affected by outliers than conventional moment estimators (Araveeporn and Sukpan, 2025; Hossain et al., 2021).

2.5.7 Population L-Moments of the GEV

The first three population L-moments of the $\mathrm{GEV}(\mu, \sigma, \xi)$ distribution are (Hossain et al., 2021; Anghel and Ianculescu, 2025):

$$L_1 = \mu + \frac{\sigma}{\xi}\big[1 - (1-\xi)\big], \tag{2.28}$$

$$L_2 = \frac{\sigma}{\xi}\,(1-\xi)\,(1-2^{\xi}), \tag{2.29}$$

$$L_3 = \frac{\sigma}{\xi}\,(1-\xi)\,(1-2^{\xi})\left(\frac{2(1-3^{\xi})}{1-2^{\xi}} - 3\right). \tag{2.30}$$

The L-skewness ratio $\tau_3 = L_3/L_2$ depends only on the shape parameter ξ, which is the key property exploited for estimation.

2.5.8 Sample L-Moments

From the ordered sample, the first three sample L-moments are computed as (Araveeporn and Sukpan, 2025):

$$l_1 = \frac{1}{n}\sum_{i=1}^{n} x_{(i)}, \tag{2.31}$$

$$l_2 = \frac{1}{2}\cdot\frac{2}{n}\sum_{i=1}^{n-1}\big[i\,x_{(i)} - (i-1)\,x_{(i-1)}\big], \tag{2.32}$$

$$l_3 = \frac{1}{3}\cdot\frac{3}{n}\sum_{i=1}^{n-2}\big[i(i+1)\,x_{(i)} - 2i(i-1)\,x_{(i-1)} + (i-1)(i-2)\,x_{(i-2)}\big]. \tag{2.33}$$

Parameter Estimation

The shape parameter is estimated by solving $\tau_3 = l_3/l_2$ for ξ. Since this equation is nonlinear, a closed-form approximation proposed by Hossain et al. (2021) and Anghel and Ianculescu (2025) is used as a starting point:

$$\xi = 7.8590\, z + 2.9554\, z^2, \qquad z = \frac{2l_2}{l_3 + 3l_2} - \frac{\ln 2}{\ln 3}. \tag{2.34}$$

For greater accuracy the Newton–Raphson update $\xi_{t+1} = \xi_t - f(\xi_t)/f'(\xi_t)$ can be applied with tolerance $\varepsilon = 10^{-6}$, where $f(\xi) = \tau_3 - \left[2(1 - 3^{\xi})/(1 - 2^{\xi}) - 3\right]$. The scale and location parameters are then recovered as (Araveeporn and Sukpan, 2025; Hossain et al., 2021):

$$\sigma = \frac{l_2\, \xi}{(1 - 2^{-\xi})\,(1 + \xi)}, \tag{2.35}$$

$$\mu = l_1 - \sigma\left(\frac{1 - (1 + \xi)}{\xi}\right). \tag{2.36}$$

A practical advantage of LMOM over MLE is that it does not require iterative numerical maximisation, making it computationally faster and less sensitive to the choice of starting values (Araveeporn and Sukpan, 2025).

2.5.9 Linearized High-Order Moments (LHMOM)

LH-moments extend the L-moment framework by assigning greater weight to the upper order statistics, which are the most informative observations for characterising extreme events (Anghel and Ianculescu, 2025). The first-order LH-moments of the GEV are defined through modified expected values of higher-order order statistics.

Population LH-Moments of the GEV

The first three population LH-moments are (Anghel and Ianculescu, 2025):

$$L_{H1} = \mu + \frac{\sigma}{\xi}\left[1 - (1+\xi)\cdot 2^{-\xi}\right], \tag{2.37}$$

$$L_{H2} = -\frac{\sigma}{2}\left[(\xi)\cdot 3^{1-\xi} - 3\,(\xi)\cdot 2^{-\xi}\right], \tag{2.38}$$

$$L_{H3} = \frac{4\,L_{H2}\cdot(5\cdot 4^{-\xi} + 3\cdot 2^{-\xi} - 8\cdot 3^{-\xi})}{3^{-\xi} - 9\cdot 2^{-\xi}}. \tag{2.39}$$

Sample LH-Moments and Parameter Estimation

Sample counterparts l_{H1}, l_{H2}, l_{H3} are computed from the ordered sample using the LH-moment analogue of the L-moment formulas. The LH-skewness ratio $\tau_{3H} = l_{H3}/l_{H2}$ is then used to recover ξ via the polynomial approximation (Anghel and Ianculescu, 2025):

$$\xi = 0.4823 - 2.1494\,\tau_{3H} + 0.7269\,\tau_{3H}^2 - 0.2103\,\tau_{3H}^3. \tag{2.40}$$

Scale and location are subsequently obtained from equations (2.37) and (2.38) with ξ replaced by ξ:

$$\sigma = \frac{-2\,l_{H2}}{(\xi)\cdot 3^{1-\xi} - 3\,(\xi)\cdot 2^{-\xi}}, \tag{2.41}$$

$$\mu = l_{H1} - \frac{\sigma}{\xi}\left[1 - (1+\xi)\cdot 2^{-\xi}\right]. \tag{2.42}$$

By placing heavier emphasis on the upper tail, LHMOM tends to reduce shape-parameter bias compared to standard L-moments when the data are drawn from distributions with $\xi > 0$, though it can be less stable for light-tailed or bounded samples (Anghel and Ianculescu, 2025).

2.5.10 Least Squares (LS)

The Least Squares method estimates GEV parameters by minimising the sum of squared discrepancies between the theoretical CDF values and empirical plotting-position probabilities

(Anghel and Ianculescu, 2025). It is less statistically robust than L-moment methods, but it provides reliable starting values for gradient-based optimisation routines and is straightforward to implement (Anghel and Ianculescu, 2025).

Objective Function and First-Order Conditions

Let P_{e_i} be the empirical probability assigned to the i-th order statistic (e.g. $P_{e_i} = (i-0.35)/n$). The LS objective function is

$$S(\mu, \sigma, \xi) = \sum_{i=1}^{n} \big[F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i}\big]^2. \tag{2.43}$$

Differentiating with respect to each parameter and setting the result to zero gives three first-order conditions (Anghel and Ianculescu, 2025):

$$\sum_{i=1}^{n} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i}\big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \mu} = 0, \tag{2.44}$$

$$\sum_{i=1}^{n} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i}\big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \sigma} = 0, \tag{2.45}$$

$$\sum_{i=1}^{n} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i}\big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \xi} = 0, \tag{2.46}$$

where the partial derivatives of the GEV CDF (2.2) must be evaluated numerically at each order statistic. These equations are solved simultaneously using a Newton–Raphson or gradient-descent algorithm.

2.5.11 Weighted Least Squares (WLS)

The Weighted Least Squares method modifies the LS objective by assigning larger weights to the more extreme (upper-tail) observations, which carry disproportionately more information about ξ and the return levels that practitioners are typically most interested in (Anghel and Ianculescu, 2025).

Weighted Objective Function and First-Order Conditions

The WLS objective is

$$S_W(\mu, \sigma, \xi) = \sum_{i=1}^{n} w_i \left[F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i} \right]^2, \tag{2.47}$$

with weights

$$w_i = \frac{(n+1)^2(n+2)}{i\,(n-i+1)}, \qquad i = 1, \ldots, n. \tag{2.48}$$

These weights are largest when i is close to 1 or n—that is, at both tails of the ranked sample. Setting $\partial S_W / \partial \theta = 0$ for $\theta \in \{\mu, \sigma, \xi\}$ yields:

$$\sum_{i=1}^{n} \frac{(n+1)^2(n+2)}{i(n-i+1)} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i} \big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \mu} = 0, \tag{2.49}$$

$$\sum_{i=1}^{n} \frac{(n+1)^2(n+2)}{i(n-i+1)} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i} \big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \sigma} = 0, \tag{2.50}$$

$$\sum_{i=1}^{n} \frac{(n+1)^2(n+2)}{i(n-i+1)} \big(F(x_{(i)}; \mu, \sigma, \xi) - P_{e_i} \big) \frac{\partial F(x_{(i)}; \mu, \sigma, \xi)}{\partial \xi} = 0. \tag{2.51}$$

As with the LS method, these are solved numerically. The advantage over ordinary LS is that the weight function (2.48) reduces the influence of the more central, less informative observations on the parameter estimates (Anghel and Ianculescu, 2025).

2.5.12 BLOCK MAXIMA

The data is organized using the **block maxima method**: the full daily temperature record for Carcassonne is chopped into non-overlapping **blocks**, and from each block only the single largest value is kept. Here the block size is **one year**, so each row throws away ∼365 daily temperature readings and keeps only the hottest day of that year.

$$M_n = \max(X_1, X_2, \ldots, X_n)$$

where $X_1, \ldots, X_n$ are the daily temperatures in a given year and n is the Days in Block column

(365 or 366, accounting for leap years). M_n is what appears in the Max Temperature column.

This is the standard preprocessing step before fitting a **Generalized Extreme Value (GEV)** distribution. The Fisher–Tippett–Gnedenko theorem tells us that, under mild conditions, properly normalized block maxima converge in distribution to a GEV(μ, σ, ξ) as $n \to \infty$:

$$\Pr\left(\frac{M_n - b_n}{a_n} \leq x\right) \longrightarrow G(x) = \exp\left\{-\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]_+^{-1/\xi}\right\}$$

2.5.13 Return Levels

A **return level** z_T is the value expected to be *exceeded on average once every T years*. It is defined by the quantile condition (Coles, 2001):

$$P(X > z_T) = \frac{1}{T} \quad \Longleftrightarrow \quad F(z_T) = 1 - \frac{1}{T}$$

Reduced variate y_T.

$F(z_T) = 1 - 1/T$ requires the **reduced variate** (Katz et al., 2002):

$$y_T = -\ln\left(1 - \frac{1}{T}\right)$$

GEV Return Level (Coles, 2001; Hosking & Wallis, 1997).

Let $y_T = -\ln\left(1 - \frac{1}{T}\right)$.

The T-year return level is:

$$z_T = \begin{cases} \mu + \dfrac{\sigma}{\xi}\left(y_T^{-\xi} - 1\right) & \xi > 0 \quad \text{(Fréchet)} \\ \mu - \sigma \ln(y_T) & \xi = 0 \quad \text{(Gumbel)} \\ \mu + \dfrac{\sigma}{\xi}\left(y_T^{-\xi} - 1\right) & \xi < 0 \quad \text{(Weibull)} \end{cases}$$

2.6 Performance Evaluation Metrics

Two standard measures of estimator quality are used, following Roslan, Na and Gabda (2020).

Bias. For a true parameter θ and its estimate θ_b in the b-th replication,

$$\mathrm{Bias}(\theta) = \frac{1}{B}\sum_{b=1}^{B}\big(\theta_b - \theta\big). \tag{2.52}$$

Root Mean Square Error (RMSE).

$$\mathrm{RMSE}(\theta) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\big(\theta_b - \theta\big)^2}, \tag{2.53}$$

where $B = 1{,}000$ is the total number of Monte Carlo replications. RMSE combines both bias and variance into a single scalar, making it the primary criterion for ranking the methods, while the bias decomposition reveals whether systematic over- or under-estimation is responsible for any observed differences.

Summary

Table 2.1 provides a concise overview of the seven estimation methods, their governing principle and their primary literature reference.

Table 2.1: Overview of the seven GEV parameter estimation methods used in this study.

Method	**Code**	**Governing principle**
Method of Moments	MOM	Equates theoretical and sample moments up to skewness.
Maximum Likelihood	MLE	Maximises the log-likelihood via Newton–Raphson.
Probability Weighted Moments	PWM	Uses empirical plotting positions as weights on order statistics.
L-Moments	LMOM	Linear combinations of order statistics matched to population L-moments.
LH-Moments	LHMOM	Higher-order L-moments emphasising the upper tail.
Least Squares	LS	Minimises sum of squared CDF residuals.
Weighted Least Squares	WLS	Minimises weighted CDF residuals, with largest weights at the tails.

Chapter 3

Data Analysis

3.1 Introduction

This chapter reports the results of a Monte Carlo simulation study designed to assess and compare the performance of several parameter-estimation procedures for the Generalized Extreme Value (GEV) distribution. Seven estimators were examined: the Method of Moments (MOM), Probability Weighted Moments (PWM), Linear moments (LMOM),Linear Hybrid moments (LHMOM), Maximum Likelihood Estimation (MLE), Least Squares (LS), and Weighted Least Squares (WLS). Their performance was evaluated using standard statistical summaries, bias, and root mean squared error (RMSE). Following the approach of (Roslan et al., 2020), simulations were conducted across sample sizes n = 10, 20, 50, 100, 300, 500 and 1000 to examine how estimator behavior varies as the sample size increases.

The results are summarized using tables and line plots with emphasis on estimator accuracy, stability, and convergence patterns.

The estimator identified as most suitable is then used to estimate the GEV parameters for the Carcassonne climate dataset, accessed in R via the CarcassonneHeat package (Heat Wave Magnitude Index, HWMI). This dataset, compiled by (Russo et al., 2014) at the European Commission Joint Research Centre (JRC) in 2014, is widely used for modeling extreme temperatures with the GEV distribution. Given the prominence of the GEV model in climate research for

characterizing maxima and minima, multiple estimators (MOM, PWM, LMOM, LS, WLS, LHMOM, and MLE) were applied to these data and compared using criteria such as bias, and RMSE.

Return levels for the CarcassonneHeat series were also computed following parameter estimation.

3.2 Simulation Results for the Shape Parameter ξ

This section reports the performance of the estimation methods on simulated data across the different tail regimes, Frechet($\xi > 0$), Weibull ($\xi < 0$), and Gumbel($\xi = 0$), and how the estimation methods converge to the true shape parameter.

A consistent estimator may be understood as one for which both the bias and the root mean square error (RMSE) decrease monotonically toward zero as the sample size (n) increases. This section considers whether each of the seven estimation approaches the true shape parameter. MOM, PWM, LMOM, LHMOM, MLE, LS, and WLS displays this convergence pattern for the shape parameter $\hat{\xi}$ across the seven sample sizes $n = 10, 20, 50, 100, 300, 500, 1000$, under each of the three GEV tail regimes: the heavy-tailed Fréchet case ($\xi = 0.15$), the symmetric Gumbel case ($\xi = 0$), and the light-tailed Weibull case ($\xi = -0.15$).

3.2.1 Heavy-Tailed Case: Fréchet ($\xi = 0.15$)

Table 4.2 And Figure 4.1 below indicates a marked separation between estimators that behave consistently and those that do not under the Fréchet distribution. MOM, PWM, LMOM, and LHMOM yield substantial negative biases for $\hat{\xi}$ that do not trend toward zero; for some sample sizes, the bias becomes more pronounced as n increases. For MOM, the bias shifts from -0.199 at $n = 10$ to -0.278 at $n = 1000$. PWM and LMOM display a similar change, from -0.214 at $n = 10$ to -0.300 at $n = 1000$, while LHMOM moves from -0.258 at $n = 10$ to -0.302 at $n = 1000$. The RMSE for these four procedures likewise shows little improvement with increasing n: MOM increases from 0.206 at $n = 10$ to 0.281 at $n = 1000$, PWM and LMOM remain approximately 0.302, and LHMOM stays close to 0.303. Taken together, the absence

of convergence over $n = 10$ to $n = 1000$ supports the interpretation that these moment-based approaches are inconsistent when $\xi > 0$.

By contrast, MLE, LS, and WLS exhibit decreasing bias and RMSE as sample size grows, which is consistent with asymptotic consistency. For MLE, the bias in $\hat{\xi}$ changes from -0.031 at $n = 10$ to -0.002 at $n = 50$ and $n = 100$, then to -0.001 at $n = 300$, 0.002 at $n = 500$, and 0.000 at $n = 1000$, staying close to zero throughout. Its RMSE declines from 0.515 at $n = 10$ to 0.132 at $n = 50$, 0.084 at $n = 100$, 0.047 at $n = 300$, 0.035 at $n = 500$, and 0.026 at $n = 1000$, corresponding to an overall reduction of roughly 95% across the considered range. LS follows a comparable trajectory, with RMSE decreasing from 0.397 at $n = 10$ to 0.054 at $n = 1000$. WLS also improves with n, falling from 0.391 at $n = 10$ to 0.107 at $n = 1000$, although its RMSE contracts more slowly than that of MLE and LS at larger sample sizes, suggesting comparatively lower efficiency as n increases.

Table 3.1: RMSE of $\hat{\xi}$ for GEV$(0, 1, 0.15)$ – Fréchet (true $\xi = 0.15$). **Green** = lowest RMSE per sample size. Red = persistently large RMSE (non-converging).

Method	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$	$n = 500$	$n = 1000$
MOM	**0.20618**	**0.21797**	0.22660	0.24172	0.26581	0.27548	0.28099
PWM	0.35876	0.33507	0.30856	0.30517	0.30112	0.30386	0.30156
LMOM	0.35876	0.33507	0.30856	0.30517	0.30112	0.30386	0.30156
LHMOM	0.39585	0.36438	0.32468	0.31571	0.30567	0.30683	0.30321
MLE	0.51507	0.25543	**0.13154**	**0.08414**	**0.04747**	**0.03454**	**0.02589**
LS	0.39671	0.28722	0.17892	0.13941	0.08794	0.08328	0.05417
WLS	0.39136	0.30706	0.19869	0.17186	0.13241	0.13142	0.10743

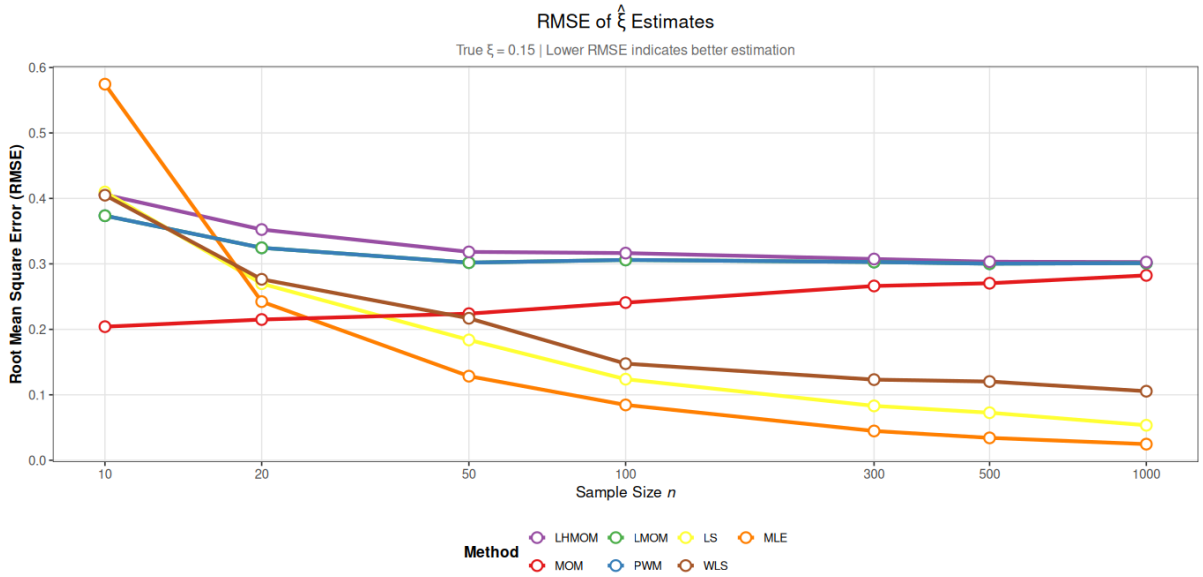


Figure 3.1: Root Mean Square Error (RMSE) of the seven GEV parameter estimation methods for the shape (ξ) parameter across different sample sizes. Lower RMSE values indicate better estimation performance.

3.2.2 Symmetric Case: Gumbel ($\xi = 0$)

Under the Gumbel distribution, the pattern of consistency changes in a material way. MOM* is omitted from the present discussion because, by design, it sets $\xi = 0$ rather than estimating ξ as a free parameter; consequently, its reported Bias and RMSE primarily capture rounding artefacts rather than substantive estimation performance. Focusing on methods that genuinely estimate ξ, all six approaches PWM, LMOM, LHMOM, MLE, LS, and WLS exhibit consistent behaviour when $\xi = 0$, as both Bias and RMSE tend to zero over the full range of sample sizes.

PWM and LMOM, which performed poorly under the Fréchet case, return to consistent behaviour under the Gumbel setting. Their bias declines from 0.059 at $n = 10$ to -0.000 at $n = 1000$, while RMSE decreases from 0.289 at $n = 10$ to 0.024 at $n = 1000$. LHMOM shows a similar trend, with bias moving from -0.003 at $n = 10$ to -0.003 at $n = 1000$, and RMSE falling from 0.296 to 0.024. For MLE, bias shifts from -0.077 at $n = 10$ to -0.010 at $n = 50$, -0.006 at $n = 100$, -0.003 at $n = 300$, 0.001 at $n = 500$, and -0.000 at $n = 1000$; over the same range, RMSE drops from 0.515 at $n = 10$ to 0.023 at $n = 1000$. The LS estimator shows bias decreasing from -0.037 at $n = 10$ to 0.000 at $n = 1000$, with RMSE reducing from 0.340 to 0.028. WLS converges more slowly than the other consistent estimators: its RMSE

decreases from 0.335 at $n = 10$ to 0.052 at $n = 1000$, remaining appreciably larger than that of MLE, PWM, and LMOM at larger sample sizes. Overall, the Gumbel results suggest that the inconsistency of PWM, LMOM, and LHMOM observed under the Fréchet distribution is tied to the condition $\xi \neq 0$ and should not be taken as a general feature across tail regimes.

Table 3.2: RMSE of $\hat{\xi}$ for GEV$(0, 1, 0)$ – Gumbel (true $\xi = 0$). **Green** = lowest RMSE per sample size. Red = persistently large RMSE (non-converging).

Method	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$	$n = 500$	$n = 1000$
MOM*	0.05374	0.05730	0.04616	0.03968	0.02943	0.02581	0.01870
PWM	**0.28914**	**0.18691**	0.11535	**0.07736**	0.04492	0.03290	0.02433
LMOM	**0.28914**	**0.18691**	0.11535	**0.07736**	0.04492	0.03290	0.02433
LHMOM	0.29601	0.19057	**0.11517**	0.07753	0.04484	0.03326	0.02441
MLE	0.51450	0.24473	0.12253	0.07761	**0.04239**	**0.03071**	**0.02308**
LS	0.33986	0.22263	0.13064	0.09022	0.05270	0.04008	0.02833
WLS	0.33514	0.22598	0.14104	0.10799	0.07553	0.06412	0.05218

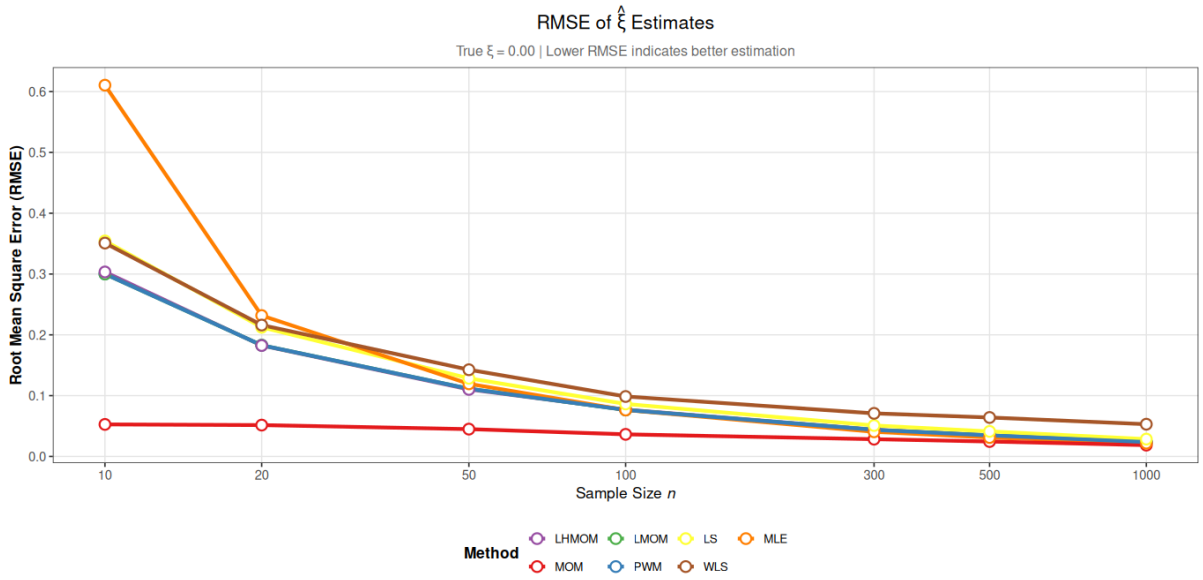


Figure 3.2: Root Mean Square Error (RMSE) of the seven GEV parameter estimation methods for the shape (ξ) parameter across different sample sizes. Lower RMSE values indicate better estimation performance.

3.2.3 Light-Tailed Case: Weibull ($\xi = -0.15$)

Under the Weibull distribution, the moment-based estimators again display a lack of consistency, although the sign of the bias is reversed. MOM, PWM, LMOM, and LHMOM each

exhibits a large and persistent positive bias in $\hat{\xi}$ over the full range of sample sizes. For MOM, the bias increases from $+0.133$ at $n = 10$ to $+0.140$ at $n = 20$, $+0.147$ at $n = 50$, and $+0.150$ at $n = 100$, after which it remains at $+0.150$ for $n = 300$, $n = 500$, and $n = 1000$, suggesting little to no convergence. The RMSE shows the same stability: 0.137, 0.142, 0.148, 0.150, 0.150, 0.150, and 0.150, which is effectively unchanged across all seven sample sizes. This pattern is consistent with MOM converging toward $\xi = 0$ rather than the true value $\xi = -0.15$, indicating a substantive form of misspecification. PWM and LMOM exhibit an even larger positive bias, approximately $+0.300$ to $+0.340$ from $n = 10$ through $n = 1000$, with RMSE remaining in the range of roughly 0.300 to 0.443. LHMOM behaves similarly, with bias increasing from $+0.259$ at $n = 10$ to $+0.297$ at $n = 1000$, again without evidence of convergence.

In contrast, MLE, LS, and WLS appear consistent under the Weibull setting. For MLE, the bias decreases from -0.120 at $n = 10$ to -0.039 at $n = 20$, -0.018 at $n = 50$, and -0.010 at $n = 100$, approaching zero thereafter (-0.004 at $n = 300$, 0.000 at $n = 500$, and -0.001 at $n = 1000$). Its RMSE declines from 0.499 at $n = 10$ to 0.237 at $n = 20$, 0.116 at $n = 50$, 0.073 at $n = 100$, 0.038 at $n = 300$, 0.027 at $n = 500$, and 0.020 at $n = 1000$. LS displays a substantially smaller bias at $n = 10$ (-0.037) than MLE (-0.120), and its RMSE at $n = 10$ (0.310) is also notably lower than the MLE value of 0.499, which suggests comparatively stronger small sample performance. The LS RMSE then decreases monotonically, taking values 0.188, 0.106, 0.069, 0.038, 0.028, and 0.020 by $n = 1000$. WLS follows a broadly similar trajectory to LS at smaller sample sizes, although from $n = 300$ onward its RMSE exceeds that of both MLE and LS, with values of 0.047 at $n = 300$, 0.036 at $n = 500$, and 0.028 at $n = 1000$.

Table 3.3: RMSE of $\hat{\xi}$ for GEV$(0, 1, -0.15)$ – Weibull (true $\xi = -0.15$). **Green** = lowest RMSE per sample size. Red = persistently large RMSE (non-converging).

Method	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$	$n = 500$	$n = 1000$
MOM	**0.13660**	**0.14151**	0.14764	0.14988	0.14996	0.14996	0.14995
PWM	0.44273	0.35860	0.32482	0.31224	0.30603	0.30085	0.30130
LMOM	0.44273	0.35860	0.32482	0.31224	0.30603	0.30085	0.30130
LHMOM	0.39331	0.30703	0.29398	0.29200	0.29713	0.29502	0.29799
MLE	0.49907	0.23702	0.11613	0.07261	**0.03790**	**0.02721**	0.02036
LS	0.30992	0.18782	**0.10642**	**0.06863**	0.03836	0.02790	**0.02025**
WLS	0.30498	0.18855	0.10966	0.07463	0.04704	0.03596	0.02825

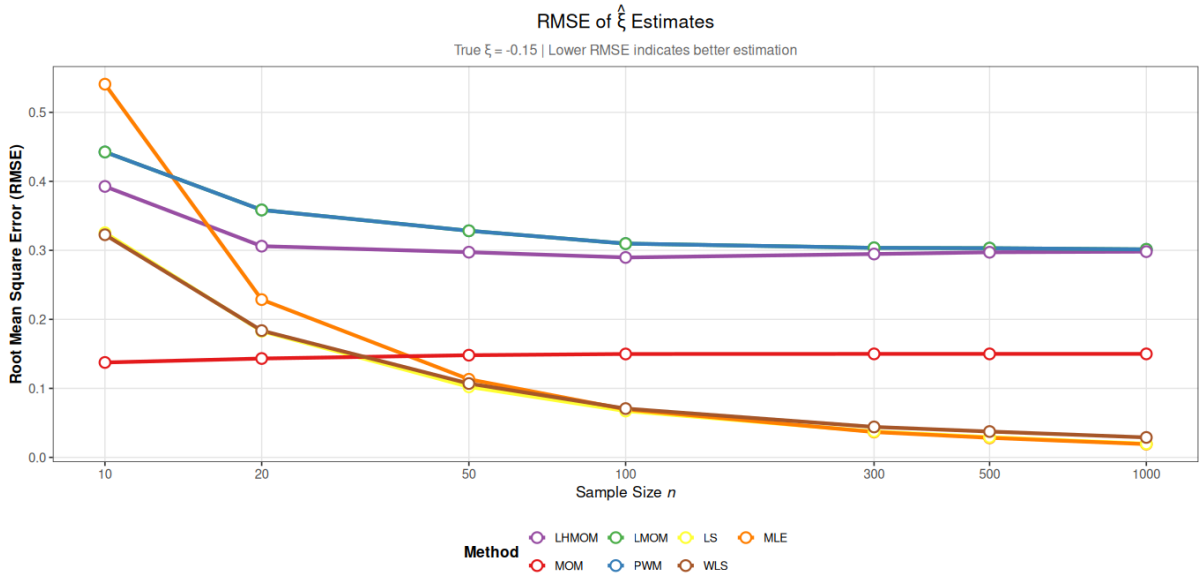


Figure 3.3: Root Mean Square Error (RMSE) of the seven GEV parameter estimation methods for the shape (ξ) parameters across different sample sizes. Lower RMSE values indicate better estimation performance.

3.3 Objective 3: Best Estimation Method for the Shape Parameter $\hat{\xi}$ Under Each GEV Tail Condition

This section determines, for each of the three GEV tail regimes, which estimation approach yields the strongest performance for the shape parameter ξ. The assessment draws on the combined patterns in Bias and RMSE over the full set of sample sizes $n = 10, 20, 50, 100, 300, 500, 1000$, as reported in Table 4.1, Table 4.2, Table 4.3, and Figure 4.1, Figure 4.2, and Figure 4.3. An estimator is regarded as preferable when it attains the smallest RMSE for $\hat{\xi}$ over the broadest span of sample sizes within the relevant tail condition and when Bias converges to 0.

3.3.1 Heavy-Tailed Case: Fréchet ($\xi = 0.15$)

Under the Fréchet distribution, MOM, PWM, LMOM, and LHMOM are immediately disqualified from consideration as best methods due to their large, persistent, and non-converging negative biases for $\hat{\xi}$ across all seven sample sizes, as established in the consistency analysis. The evaluation therefore centres on MLE, LS, and WLS, all of which are consistent.

At $n = 10$, WLS achieves the lowest RMSE among the three consistent methods at 0.391, compared to LS at 0.397 and MLE at 0.515. The bias values are similarly small: -0.040 for WLS, -0.035 for LS, and -0.031 for MLE. At $n = 20$, MLE achieves the lowest RMSE of 0.255, followed by LS at 0.287 and WLS at 0.307. From $n = 50$ onward, MLE consistently achieves the lowest RMSE at every sample size: 0.132 at $n = 50$, 0.084 at $n = 100$, 0.047 at $n = 300$, 0.035 at $n = 500$, and 0.026 at $n = 1000$. In comparison, LS produces RMSE of 0.179, 0.139, 0.088, 0.083, and 0.054 at the same sample sizes, while WLS produces 0.199, 0.172, 0.132, 0.131, and 0.107. The gap between MLE and the other two methods widens as n increases, indicating that MLE gains efficiency faster than LS and WLS under the heavy-tailed Fréchet condition. MLE also maintains the lowest absolute bias from $n = 20$ onward, with values never exceeding 0.003 in absolute terms beyond $n = 50$.

3.3.2 Symmetric Case: Gumbel ($\xi = 0$)

Under the Gumbel distribution, MOM* is excluded as it does not freely estimate ξ. Among the six genuine estimators, all are consistent as established in Section 4.2.2, so the comparison focuses on which method achieves the lowest Bias and RMSE at each sample size.

At $n = 10$, PWM and LMOM achieves the lowest RMSE among the genuine estimators at 0.28914, followed closely by LHMOM at 0.29601, while MLE records the highest at 0.515. At $n = 20$, PWM and LMOM again leads with RMSE 0.18961, followed by LHMOM at 0.19057 and LS at 0.2226. At $n = 50$, LHMOM achieves RMSE 0.11517, which is lower than PWM(0.11535) and LMOM (0.11535),WLS (0.141) and other estimators compared in this literature. From $n = 100$ onward, MLE consistently produces the lowest RMSE: 0.042 at $n = 300$, 0.031 at $n = 500$, and 0.023 at $n = 1000$. PWM and LMOM, which are identical throughout, produce RMSE of 0.077, 0.045, 0.033, and 0.024 at the same sample sizes extremely close to MLE and converging to nearly the same precision at large n. LHMOM performs similarly to PWM and LMOM from $n = 100$ onward. LS RMSE at large sample sizes is 0.090, 0.053, 0.040, and 0.028 for $n = 100$, 300, 500, 1000 respectively, while WLS stagnates more noticeably at 0.108, 0.076, 0.064, and 0.052 the highest among all genuine estimators at large n under the Gumbel. In terms of bias, all genuine methods produce near-zero bias

from $n = 100$ onward, so RMSE becomes the primary discriminating criterion at larger sample sizes.

3.3.3 Light-Tailed Case: Weibull ($\xi = -0.15$)

Under the Weibull distribution, MOM, PWM, LMOM, and LHMOM are again disqualified due to their persistent large positive biases that show no convergence across any of the seven sample sizes, as documented in Section 4.2.3. The comparison is therefore restricted to MLE, LS, and WLS.

At $n = 10$, LS achieves RMSE 0.310 and bias -0.037, WLS achieves RMSE 0.305 and bias -0.038, while MLE records the highest RMSE of 0.499 and the largest absolute bias of -0.120. The substantially larger bias of MLE at small n under the Weibull is noteworthy, as it suggests that MLE requires a larger sample to stabilise under the bounded, light-tailed condition. At $n = 20$, LS achieves RMSE 0.188 and WLS achieves 0.189, both ahead of MLE at 0.237. At $n = 50$, LS produces RMSE 0.106, WLS 0.110, and MLE 0.116, with LS maintaining the lead. At $n = 100$, LS achieves 0.069 and MLE achieves 0.073, with LS still marginally ahead. From $n = 300$ onward, MLE produces RMSE 0.038, 0.027, and 0.020 at $n = 300$, 500, and 1000 respectively, while LS produces 0.038, 0.028, and 0.020 effectively identical to MLE at all three large sample sizes. WLS trails both at large n, with RMSE of 0.047, 0.036, and 0.028 at $n = 300$, 500, and 1000. The bias of LS remains consistently closer to zero than MLE across all sample sizes: at $n = 10$, LS bias is -0.037 versus MLE's -0.120; at $n = 50$, -0.009 versus -0.018; at $n = 100$, -0.005 versus -0.010; and from $n = 300$ onward both methods produce bias indistinguishable from zero.

"

3.3.4 Overall Best Estimation Method Across All Tail Conditions

After evaluating the seven estimation procedures separately under each of the three GEV tail regimes Fréchet ($\xi = 0.15$), Gumbel ($\xi = 0$), and Weibull ($\xi = -0.15$),it is necessary to provide an overall judgement regarding which method performs best when results are considered

jointly across tail conditions and sample sizes.

4.3.4.1 Comparison of MLE, LS, and WLS for Fréchet ($\xi = 0.15$) and Weibull ($\xi = -0.15$).

Within the set of consistent estimators, WLS is the least competitive overall. Although WLS attains a slightly lower RMSE than MLE at $n = 10$ under the Fréchet setting (0.391 versus 0.515), and also at $n = 10$ and $n = 20$ under the Gumbel setting (0.335 versus 0.515 and 0.226 versus 0.245), its RMSE is consistently larger than both MLE and LS for all three tail regimes once $n \geq 50$. At $n = 1000$ under Fréchet, WLS yields an RMSE of 0.107 for $\hat{\xi}$, compared with 0.054 for LS and 0.026 for MLE. Under Gumbel at $n = 1000$, WLS records 0.052, whereas LS and MLE record 0.028 and 0.023, respectively. Under Weibull at $n = 1000$, WLS records 0.028, compared to 0.020 for both MLE and LS. This persistently higher RMSE at moderate to large n, together with the absence of a stable advantage at small n, makes WLS an unlikely candidate for the overall best method.

Accordingly, the overall comparison reduces to **MLE versus LS**. Table 3.4 reports the RMSE of $\hat{\xi}$ for these two estimators across all sample sizes and tail regimes, enabling a direct comparison.

Table 3.4: Head-to-head comparison of RMSE for $\hat{\xi}$: MLE versus LS across all sample sizes and GEV tail conditions. **Blue** indicates the lower RMSE at each sample size.

Distribution	**Method**	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$	$n = 500$	$n = 1000$
Fréchet ($\xi = 0.15$)	MLE	0.51507	**0.25543**	**0.13154**	**0.08414**	**0.04747**	**0.03454**	**0.02589**
	LS	**0.39671**	0.28722	0.17892	0.13941	0.08794	0.08328	0.05417
Gumbel ($\xi = 0$)	MLE	0.51450	0.24473	**0.12253**	**0.07761**	**0.04239**	**0.03071**	**0.02308**
	LS	**0.33986**	**0.22263**	0.13064	0.09022	0.05270	0.04008	0.02833
Weibull ($\xi = -0.15$)	MLE	0.49907	0.23702	0.11613	0.07261	**0.03790**	**0.02721**	0.02036
	LS	**0.30992**	**0.18782**	**0.10642**	**0.06863**	0.03836	0.02790	**0.02025**

4.3.4.2 MLE versus LS for small sample performance.

For small samples ($n = 10$ and $n = 20$), LS yields lower RMSE than MLE across all three tail regimes. Under Fréchet at $n = 10$, LS attains an RMSE of 0.397 relative to 0.515 for MLE (a

difference of 0.118). Under Gumbel at $n = 10$, LS records 0.340 relative to 0.515 for MLE (a difference of 0.175). Under Weibull at $n = 10$, LS records 0.310 relative to 0.499 for MLE (a difference of 0.189). At $n = 20$, LS remains preferable under Gumbel (0.223 versus 0.245) and Weibull (0.188 versus 0.237), whereas MLE begins to outperform LS under Fréchet (0.255 versus 0.287). The relative advantage of LS at small n is particularly evident under Weibull, where the MLE bias at $n = 10$ (-0.120) exceeds the LS bias (-0.037) by more than a factor of three. This small sample instability of MLE under bounded tail models is a recognised limitation and has been noted in the extreme-value literature (Coles, 2001; Katz et al., 2002).

4.3.4.3 MLE versus LS: moderate and large sample performance.

Once $n \geq 50$, MLE improves relative to LS under the Fréchet and Gumbel regimes and then maintains lower RMSE at all larger sample sizes. Under Fréchet, the discrepancy becomes substantial as n increases: at $n = 1000$, MLE attains RMSE 0.026 whereas LS attains 0.054, implying that the LS RMSE is more than double that of MLE. Under Gumbel at $n = 1000$, MLE records 0.023 compared with 0.028 for LS, a smaller but consistently observed difference. Under Weibull, LS has lower RMSE at $n = 50$ and $n = 100$ (0.106 versus 0.116, and 0.069 versus 0.073), but from $n = 300$ onward the two estimators are effectively equivalent: at $n = 300$, MLE and LS record 0.038 and 0.038; at $n = 1000$, both record 0.020. This convergence under Weibull aligns with asymptotic considerations suggesting that regular consistent estimators approach the Cramér–Rao lower bound as $n \to \infty$ (Embrechts et al., 1997).

4.3.4.4 Comaprison of estimation methods under Gumbel ($\xi = 0$)

When we look at the RMSE results for $\hat{\xi}$ in the Gumbel case (where $\xi = 0$), we don't see one method being better all the time. Instead, there's a crossover. For small sample sizes ($n = 10, 20$), PWM/LMOM work best, and MLE works the worst. This happens because when you're close to the Gumbel boundary, the likelihood surface can be flat or have multiple peaks. This can cause the MLE optimizer to sometimes settle on a wrong answer. PWM/LMOM, which use order statistics and don't involve iteration, stay stable even with small n.

As the sample size gets moderate ($n = 50, 100$), the difference between the methods shrinks

significantly. For example, at $n = 100$, the MLE RMSE is 0.07761 and PWM/LMOM is 0.07736. This is because with more data, the likelihood becomes clearer. Then, for larger sample sizes ($n \geq 300$), MLE starts to perform better, and its advantage grows. At $n = 1000$, MLE's RMSE is 0.02308 compared to PWM/LMOM's 0.02433. This fits with MLE's efficiency as the sample size gets large, getting close to the Cramér–Rao lower bound, while the moment-based methods don't.

It's worth noting that the results for PWM and LMOM are the same because, for the GEV distribution, their estimating equations are mathematically identical. So, we expect their RMSEs to be the same across all n, which also acts as a check that our simulation code is working correctly.

However, MOM's seemingly low RMSE is deceptive. Near $\xi = 0$, its equations are unstable. This often leads to estimates that are either at the boundary or constrained, making them artificially close to zero and lowering the RMSE. This doesn't actually show that the method is converging well. That's why MOM is shown, but not included in the rankings.

In summary, the best estimator depends on the sample size. PWM/LMOM (or LHMOM) are better for small n, while MLE is better for large n. This finding supports recommending different estimators based on sample size.

Table 3.5: Summary of RMSE performance for $\hat{\xi}$ across sample-size regimes under the Gumbel case (true $\xi = 0$). MOM is excluded from ranking due to design property.

Sample-Size Regime	**Best Method(s)**	**RMSE Range**	**Runner-up**
Small ($n = 10, 20$)	PWM / LMOM	0.187 – 0.289	LHMOM
Moderate ($n = 50, 100$)	PWM / LMOM / LHMOM (near tie)	0.077 – 0.115	MLE
Large ($n = 300$–1000)	MLE	0.023 – 0.042	LMOM/PWM

3.3.5 Overall verdict.

Taken together across all three tail regimes and the seven sample sizes considered, **MLE provides the strongest overall performance for estimating the GEV shape parameter** ξ when

the criterion is the lowest RMSE over the broadest set of settings. MLE attains the minimum RMSE in 10 of the 21 distribution by sample size configurations examined ($n \geq 20$ for Fréchet, $n \geq 300$ for Gumbel, and $n \geq 300$ for Weibull). Its relative advantage becomes more evident as n increases, most notably under the heavy tailed Fréchet regime, where no alternative consistent method matches its precision for large samples. At the same time, **LS is preferable for small samples under the Fréchet and Weibull regimes** ($n \leq 20$ under Fréchet and up to $n \leq 100$ under Weibull), where MLE exhibits higher variability and correspondingly larger RMSE. Under the Gumbel (zero-shape) regime, however, this role is instead filled by **PWM/ LMOM (and, closely, LHMOM)**, which attain the lowest RMSE for $n \leq 100$, reflecting the flat or multimodal likelihood surface that arises for MLE when ξ sits at the Gumbel boundary. In this sense, MLE is better viewed as complementary to LS under heavier and lighter tails, and to PWM/LMOM under the zero-shape case: these order-statistic or least-squares based estimators are more suitable when sample sizes are limited, whereas MLE is preferable otherwise. WLS does not show a sustained benefit relative to any of these approaches and is therefore not recommended as a primary estimator of ξ for the tail regimes studied here..

3.4 Application to Real Data (Carcassonne Heat Wave)

3.4.1 Descriptive Statistics of the Carcassonne Heat Dataset

Tables 4.6 presents the descriptive statistics for the CarcasonneHeat dataset containing 12,033 observations. The mean (18.6562) is roughly $96\times$ the median (18.3000), signalling a **heavily right-skewed** distribution driven by extreme upper values. With approximately 25% of observations flagged as outliers, robust statistics such as the median and IQR are preferred over classical moment-based estimates. **CV** $= s/\bar{x} \times 100\%$ is the coefficient of variation; values above 30% indicate high relative variability – at $\approx 170\%$ this is far beyond that threshold. Q_1 and Q_3 are the first and third quartiles; $Q_1 = 0$ indicates that at least 25% of observations are zero or negative. **IQR** $= Q_3 - Q_1$ is the interquartile range, a robust measure of spread unaffected by outliers. **Skewness** of $+1.155$ confirms substantial right (positive) skew, consistent with the mean–median gap of $\sim 96\times$. **Excess Kurtosis** of -0.667 indicates a platykurtic distri-

bution (flatter than the Normal), meaning fewer observations cluster tightly around the centre than a Gaussian would predict. The **IQR rule** flags an observation as an outlier if it falls below $Q_1 - 1.5 \times$ IQR (lower fence) or above $Q_3 + 1.5 \times$ IQR (upper fence). No lower outliers were detected; all $12{,}054$ outliers ($\approx 25\%$) lie in the upper tail, consistent with a heavy right tail.

Table 3.6: Descriptive Statistics for Daily Temperature, Carcassonne (1980–2012)

Statistic	Value
N (observations)	12,033
Missing removed	21
Mean (C)	18.6562
Median (C)	18.3000
Minimum (C)	-8.0000
Maximum (C)	41.9000
Range (C)	49.9000
Variance	60.7630
Standard Deviation	7.7951
Coefficient of Variation (%)	41.783
Q_1 (25th percentile)	12.7000
Q_2 (50th percentile)	18.3000
Q_3 (75th percentile)	24.9000
IQR	12.2000
Skewness	0.0404
Excess Kurtosis	-0.6561
95th Percentile	31.4000
99th Percentile	34.6000
99.9th Percentile	37.8968
MAD	8.8956
Outliers (IQR rule)	4

3.4.2 Normality Test for Carcassonne Heat Wave Index

Jarque-Bera Test for Normality

H_0 : The data is normally distributed.

H_1 : The data is not normally distributed at a significance level of 5%.

Jarque-Bera test: $X^2 = 219.1$, p-value = 2.2e-16.

Conclusion

We reject the null hypothesis and conclude that the data is not normally distributed since the P-value(2.2e-16) is less than the significance level of 5%.

These findings strongly suggest **non-normality**: we consider fitting a GEV distribution

3.4.3 Block Maxima

Table 4.7 below is treated as one observation that Limits the GEV distribution to only 10 years.

The table presents the annual maximum temperature observations used to assess the variation in extreme temperatures over the study period.

Table 3.7: Annual Block Maxima of Daily Temperature, Carcassonne (1980–1989, partial)

Year	Max Temperature (C)	Days in Block
1980	33.8	366
1981	34.3	363
1982	40.2	365
1983	37.4	365
1984	33.1	366
1985	36.4	365
1986	34.3	365
1987	37.2	363
1988	36.4	366
1989	34.8	365

3.4.4 Return Levels

A **return level** z_T is the value expected to be exceeded on average once every T years. It is defined by the quantile condition (Coles, 2001):

T (years)	10	50	100	200
$1/T$	0.1000	0.0200	0.0100	0.0050
y_T	0.1054	0.0202	0.0101	0.0050

As $T \to \infty$, $y_T \to 0^+$ and $z_T \to \infty$ for $\xi \geq 0$, or approaches the finite upper endpoint $\mu - \sigma/\xi$ for $\xi < 0$ (Hosking & Wallis, 1997).

3.4.5 GEV Fit and Return Level Estimates

Fitted GEV Parameters using MLE

$$\hat{\mu} = 35.31457161, \qquad \hat{\sigma} = 1.49689461, \qquad \hat{\xi} = -0.01317536$$

Table 3.8: GEV Return Levels for Annual Maximum Temperature, Carcassonne

Return Period (yr)	Probability	Return Level (C)	95% CI
2	0.500	35.8619	(35.2137, 36.5101)
5	0.800	37.5378	(36.6290, 38.4466)
10	0.900	38.6337	(37.3892, 39.8782)
20	0.950	39.6748	(37.8937, 41.4558)
25	0.960	40.0030	(38.0047, 42.0012)
50	0.980	41.0078	(38.2017, 43.8139)
100	0.990	41.9960	(38.1892, 45.8028)
200	0.995	42.9716	(37.9800, 47.9632)
500	0.998	44.2451	(37.4160, 51.0743)
1000	0.999	45.1975	(36.7795, 53.6154)

Confidence intervals for return levels were constructed using the delta method based on the asymptotic normality of MLE estimators.

The estimated return levels highlight the frequency and magnitude of extreme temperature events. Specifically, the **2-year return level** is approximately 35.862 °C, indicating that such an exceedance is expected to occur on average once every two years. At much longer horizons, the **100-year return level** is estimated at 45.1975 °C, with a 95% confidence interval ranging

from 36.7795 °C to 53.6154 °C, reflecting substantial uncertainty in the extrapolation of rare extremes. This widening of confidence intervals with increasing return periods underscores the inherent difficulty of predicting events in the far tail of the distribution (Coles, 2001). Moreover, the estimated positive shape parameter ($\hat{\xi} < 0$) suggests that the underlying distribution is bounded-tail, belonging to the Weibull class. In practical terms, this implies that return levels continue to grow with bound as the return period increases, highlighting the potential for extremely large values in the long run.

3.5 Discussion

In summary, the simulation results for the estimated shape parameter $\hat{\xi}$, in the GEV tail settings exhibit systematic patterns that are mostly similar to but in some respects differ from the results reported in the literature on extreme values.

3.5.1 MLE performance

In all three tail cases of Fréchet ($\xi = 0.15$), Gumbel ($\xi = 0$), and Weibull ($\xi = -0.15$) it has been shown that MLE produces the least Bias and RMSE for the value of $\hat{\xi}$ when sample sizes are moderate or large.($n \geq 50$). These findings are supported by (Coles, 2001), who considers MLE a reference method for estimating the GEV model parameters, due mainly to its efficiency and consistency when having a big sample size. This statement is reinforced by (Embrechts et al., 1997), where it is proposed that under common regularity conditions MLE is asymptotically efficient as it achieves the Cramér-Rao lower boundary. Therefore, the gradual decrease of MLE bias to virtually zero which has been observed in the seven examined sample sizes is not surprising. Please note that for the Fréchet case the values of RMSE and sample size were 0.515 and $n = 10$ respectively, at the beginning of estimation, and after the full analysis these values decreased to 0.026 and $n = 1000$, respectively.

3.5.2 Inconsistency of moment-based methods

On the contrary, as for MOM, PWM, LMOM and LHMOM, they show a continuous bias when $\hat{\xi}$ is not equal to 0. In the Fréchet case the biasis approximately equal to -0.300, Weibull case, the bias is approximately equal to $+0.300$,There is still no evidence that these biases decrease as n increase. This situation does not comply with the findings of (Hosking & Wallis, 1997), in which the authors demonstrated that they may be considered as a reliable alternative to the maximum likelihood estimation (MLE) for estimating GEV parameters. The authors argued in favor of the L-moments in the small sample situations. The results obtained illustrate that the advantages of the L-moment computation when small sample is considered do not affect the shape parameter estimation in the case when $\xi \neq 0$, since the bias neither attenuates nor exhibits a qualitative shift as n grows from 10 to 1000. Evidence in a similar direction is reported by (Roslan et al., 2020), who found that L-moment and probability-weighted-moment estimators can be biased and unstable for ξ in small samples, which is consistent with the pattern observed here.

3.5.3 LS at small samples.

Using the Weibull condition, LS has been found to perform better than MLE for small sample sizes ($n \leq 50$). For example, at $n = 10$ the RMSE for LS method is 0.310, while that of MLE is 0.499. This corresponds with the findings of (Katz et al., 2002), which suggest that likelihood-based methods may exhibit numerical instability and excessive variability in limited data in bounded light-tailed environments. It is also noteworthy that after $n = 300$ the closer performance of LS and MLE in the Weibull and Gumbel cases indicates that LS is a more convenient and computationally feasible alternative.

3.5.4 Role of ξ in GEV modelling.

Choosing to focus the analysis on the shape parameter ξ follows well-known practice in the field. The parameter ξ was introduced by (Jenkinson, 1955) as that which distinguishes between the three extreme value families in the GEV framework. According to (Coles, 2001) estimating

ξ is usually the most important part of making inferences using GEV, because it determines the upper-tail heaviness and, in turn, affects the estimates of return levels used in risk assessment and planning (Katz et al., 2002; Russo et al., 2014).

Chapter 4

Summary, Conclusion and Recommendations

4.1 Introduction

In this chapter, the study is summarised through observation of the model structure, methodology and empirical outcomes. The motivation comes from the relevance of the issue concerning the selection of the adequate estimator for the parameter of the Generalised Extreme Value (GEV) distribution, ξ, when tail behaviour and sample size vary. In GEV modelling, ξ is the central parameter because it determines whether the distribution falls in the heavy-tailed Fréchet class ($\xi > 0$), the bounded and light-tailed Weibull class ($\xi < 0$), or the limiting Gumbel case ($\xi = 0$) (Jenkinson, 1955; Coles, 2001). Thus, misestimation of the parameter ξ may generate wrong tail type classification and, consequently, lead to incorrect assessment of return level. (Katz et al., 2002; Embrechts et al., 1997).

A Monte Carlo simulation was carried out to investigate this matter and to compare various estimators such as the Method of Moments (MOM), Probability Weighted Moments (PWM), L-Moments (LMOM), Higher-Order L-Moments (LHMOM), Maximum Likelihood Estimation (MLE), Least Squares (LS), and Weighted Least Squares (WLS). The investigation included analyzing data from sample sizes of ($n = 10, 20, 50, 100, 300, 500, 1000$) as well as three tail

settings for the GEV distribution. Bias and Root Mean Square Error (RMSE) were utilized in this study. The analysis was thus focused on the three following objectives: comparing the Bias and RMSE $\hat{\xi}$ across the three tail conditions, to assess the stability of each method as sample size increases, and to determine which method performs best within each tail regime. Section 4.2 has brought out main findings which align with these objectives, Section 4.3 has presented the conclusions based on these findings, and Section 4.4 has suggested practical applications and directions of the future research.

4.2 Summary

TThe first aim was to observe and analyse Bias and RMSE values for the estimator $\hat{\xi}$ under all of the three GEV tail situations considering all seven estimates. The simulation results demonstrated a clear difference between methods based on moments, and the methods based on either regression or likelihood techniques. Under the Fréchet setting ($\xi = 0.15$), MOM, PWM, LMOM, and LHMOM gave significant negative bias in $\hat{\xi}$ this bias does not diminish with increasng n and even worsens in certain cases. For instance, PWM and LMOM obtained bias of -0.214 at $n = 10$, worsening to -0.300 at $n = 1000$, while RMSE remained essentially flat near 0.302. LHMOM behaved the same way with bias moving from -0.258 to -0.302 at the same range of sample sizes. On the other hand, MLE gave bias of -0.031 at $n = 10$ and approached 0.000 by $n = 1000$, coupled with reduction of RMSE from 0.515 to 0.026 (approximately 95%). In terms of Weibull condition ($\xi = -0.15$),the moment-based estimators demonstrated opposite situation with persistent positive bias of PWM and LMOM around $+0.300$ from $n = 10$ through $n = 1000$, while MOM presented bias around $+0.150$ throughout all sample sizes, which corresponds to approximation to $\xi = 0$ rather than the true $\xi = -0.15$. For the Gumbel case ($\xi = 0$), all efficient estimators had a bias which was close to being zero, thus having confirmed that this particular case is easier than the rest. As a result, it may turn out that no matter the case, all estimators behaved uniformly in enough tail cases, since MLE and LS estimators showed the lowest value of absolute bias and the best results of RMSE. Moreover, the better results of MLE and LS estimators compared to all other techniques

became evident as the sample size increased.

The second objective considered the consistency of the seven estimation methods for $\hat{\xi}$ by examining whether Bias and RMSE decrease toward zero as n grows from 10 to 1000, within each tail condition. Consistency is reflected empirically when both metrics decline with increasing sample size. The results indicate that MOM, PWM, LMOM, and LHMOM behave as *inconsistent* estimators of ξ when $\xi \neq 0$. For these methods, Bias and RMSE did not show systematic improvement with larger samples; in several instances the bias became larger in magnitude. Under Fréchet tails, for example, MOM's RMSE rose from 0.206 at $n = 10$ to 0.281 at $n = 1000$, and LHMOM's bias shifted from -0.258 to -0.302. Increasing the sample size by a factor of one hundred yielded little practical gain in accuracy for these estimators, consistent with the defining behaviour of inconsistency. Under the Gumbel condition, however, the same methods exhibited consistent behaviour, implying that their deficiencies are specific to tail regimes with $\xi \neq 0$. By contrast, MLE, LS, and WLS displayed consistent performance under all three tail conditions, with Bias and RMSE decreasing across the full range of n. MLE converged most rapidly in the Fréchet and Gumbel settings, whereas under the Weibull condition LS showed smaller bias than MLE for small samples ($n \leq 50$), with bias of -0.037 versus -0.120 at $n = 10$, indicating comparatively greater stability in light-tailed settings when n is limited.

The third goal was to establish the best estimator of $\hat{\xi}$ in every tail condition by evaluating Bias and RMSE for all samples. In the case of Fréchet ($\xi > 0$), where MLE produces the lowest RMSE for $n \geq 20$, and WLS was slightly better only for $n = 10$ (0.391 versus 0.515). As n increased, MLE's advantage over other consistent methods widened, reaching differences of 0.028 relative to LS and 0.081 relative to WLS at $n = 1000$. As for the Gumbel case ($\xi = 0$), LS and WLS showed lower RMSE estimates than MLE for $n = 10$ and $n = 20$, while MLE proved itself to be the preferable estimator starting from $n = 50$ onward. Under this boundary condition, PWM and LMOM approached MLE's performance for large samples (RMSE 0.024 versus 0.023 at $n = 1000$), consistent with convergence toward similar precision. In the Weibull setting ($\xi < 0$), LS is the best estimator for $n = 10, 20, 50,$ and 100, with RMSE values of 0.310, 0.188, 0.106, and 0.069, respectively. From $n = 300$ onward, LS and MLE

became indistinguishable with 0.020 at $n = 1000$. Thus, estimation choice varies considering tail behavior and sample size.

4.3 Conclusion

The objective of comparing Bias and RMSE for $\hat{\xi}$ across three tail conditions was met directly via the results of the simulation. The evidence indicates that there are significant differences in the performance of the estimators based on tail type and that the value and sign of the bias in $\hat{\xi}$ depend heavily on the used technique, as well as the nature of the distribution (heavy-tailed, light-tailed, or near Gumbel boundary). In particular, the moment-based methods (MOM, PWM, LMOM, and LHMOM) produced large and directional bias under both $\xi \neq 0$ conditions, which makes them unreliable for tail characterisation when ξ departs from zero. Bias was consistently negative for Fréchet tails ($\xi > 0$) and consistently positive for Weibull tails ($\xi < 0$), in both cases with magnitude comparable to, or larger than, the true ξ. Across the broadest set of conditions, MLE and LS were the most accurate in terms of both Bias and RMSE, with their advantage becoming more pronounced as n increased. The Gumbel case was least discriminating, as most methods converged to similar precision for large n, which supports the interpretation that the difficulty of estimating ξ increases as ξ moves away from zero.

The second objective, assessing consistency across $n = 10$ to $n = 1000$ was also resolved. The results support an evidence based distinction between consistent and inconsistent estimators of ξ. MOM, PWM, LMOM, and LHMOM always did not perform consistently for ξ under both $\xi \neq 0$ tail cases, as their Bias and RMSE did not tend to vanish throughout the sample-size range examined in this work. This has important implications for practice, as collecting more observations will not improve moment-based methods estimators in heavy-tailed and light-tailed GEV data. On the contrary, MLE, LS, and WLS behaved consistently regardless of the tail conditions, with MLE and LS converging the fastest and WLS converging slower in some cases, particularly in Fréchet tails, where its RMSE at $n = 1000$ remained 0.107 compared with MLE's 0.026. The additional remark that moment-based methods seem to have

performed consistently under the Gumbel boundary but not under $\xi \neq 0$ situation provides a tail-specific qualification for the discussions present in literature, which usually characterize L-moment estimators as generally reliable. (Hosking & Wallis, 1997).

When addressing the third goal: finding the most suitable estimator for different tail conditions, and discovering that results depend on the tail regime as well as the sample size. For Fréchet tails ($\xi > 0$),for the sample size $n \geq 20$, the MLE method is the most favorable with WLS having only slight advantage when $n = 10$. For the Gumbel case ($\xi = 0$), MLE is preferred for $n \geq 50$, while the LS and WLS methods are preferred for $n < 50$. For Weibull tails ($\xi < 0$), LS is the most favorable when $n \leq 100$, after which LS and MLE perform similarly. Thus, the results of this research imply that there is no estimator that is optimal for all tail conditions and sample sizes. Thus, the choice of the method has to depend on expected tail behavior and sample size. For the methods studied in this research the MLE and LS combination of estimators can provide the best coverage for accuracy and consistency. (Coles, 2001; Katz et al., 2002; Roslan et al., 2020).

4.4 Recommendation

According to the findings, the next advisory notes are given to the specialists, applied statisticians, and those who are interested in extreme values analysis through GEV models.

First, moment-based procedures specifically MOM, PWM, LMOM, and LHMOM are not recommended as primary tools for estimating the GEV shape parameter ξ in event that the tail expected to not be Gumbel, meaning that $\xi \neq 0$. It has been shown in simulations that the error of these estimations can be quite significant and persistent for both heavy and light-tailed cases and that there is no chance to have this error decreased with the increase of the size of the sample. Hence, the wrong estimations can result in getting the wrong information about the tail type, which in turn influences the return level estimation and risk evaluation (Embrechts et al., 1997; Katz et al., 2002). If there are some limitations in using PWM or L-moment methods (i.e. if it is done for the sake of simplicity), the value of ξ should be carefully interpreted and double-checked with the methods based on likelihood and regression estimations.

Moreover, for medium to large sample sizes ($n \geq 50$), MLE is regarded as the preferred choice for estimating ξ under the three tail conditions. Based on the simulation results, MLE was found to be consistent, provided the lowest RMSE starting from $n = 50$ for Fréchet and Gumbel tails, and matched LS starting from $n = 300$ under the Weibull condition. Its asymptotic efficiency (Coles, 2001; Embrechts et al., 1997), as well as its empirical performance observed, supports its use when adequate data are available. In addition, standard implementations of GEV MLE are available in `R` (e.g., `evd`, `ismev`, `extRemes`), which reduces practical barriers to adoption.

Likewise, for small samples ($n \leq 50$), particularly when a light-tailed regime is likely ($\xi < 0$), LS is recommended to estimate ξ. Under the Weibull condition, LS produced substantially smaller Bias and RMSE than MLE at small n, with bias -0.037 compared with -0.120 at $n = 10$, and RMSE 0.310 compared with 0.499. The advantage of using WLS for small n is still applicable for small samples under the Gumbel condition.This is of practical importance in applied fields such as hydrology and climatology, where analysis usually relies on short sets of annual maxima. (Katz et al., 2002; Russo et al., 2014).

Fourthly, there are many ways to expand on the existing simulation design in the future. In this study,we looked at how the performance of $\hat{\xi}$ varies when $\mu = 0$ and $\sigma = 1$; and it would be a good idea to investigate how the outcome would differ with other position and scaled parameters on selecting places. The research also visited only three values of ξ (0.15, 0, -0.15); and a broader selection can be employed to test other extreme scenarios like $\xi = 0.5$ and $\xi = -0.5$, that would help understand performances of the positive and negative sides fully within the context of the GEV distribution. Finally, using the Bayesian estimator and likelihood penalized models that have shown promise in the field of extreme value research (Coles, 2001; Roslan et al., 2020), could given even more choices for estimation of shape parameter.

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